Exercise 1: Worked solutions

Mathematics for Engineers I · Module 62765 · Exercise Sheet 1

From Fundamentals to AI Apocalypse · personal study guide

Contents

Exercise 1.1: Statements 2

Exercise 1.2: Sets 3

Exercise 1.3: Quantifiers — students and courses 4

Self-Study 1.4: Statements 5

Self-Study 1.5: Sets 6

Self-Study 1.6: Quantifiers and numbers 6

Answers 7

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Try each part before opening its solution. The worked steps show how to translate the wording into logic, test membership one element at a time, and keep quantifier order straight.

Exercise 1.1: Statements

A logical statement is a sentence with one definite truth value. A sentence can be a statement even when we do not know whether it is true; an unassigned variable or an undefined judgement can prevent a definite truth value.

Connected topic: Propositional logic, predicates and quantifiers (§1.1) define statements, open forms and truth values.

1a: Classify the four expressions

Try first: for each sentence, ask whether its variables and context are fixed well enough that it must be true or false.

Answers: see "Exercise 1.1: Statements" under Answers at the end.

1b: Compare three equivalent forms

Try first: make a row for each truth-value pair of A and B, then calculate all three expressions independently.

Answers: see "Exercise 1.1: Statements" under Answers at the end.

1c: Find the guilty suspect

Try first: write each clue as an implication, then test whether Christoph can be guilty.

Answers: see "Exercise 1.1: Statements" under Answers at the end.

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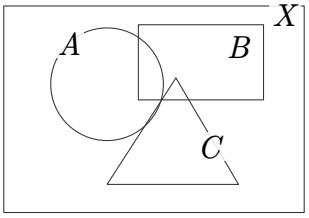
Exercise 1.2: Sets

Translate set operations into membership conditions. This keeps the meaning precise even when the sets overlap in a complicated way.

Connected topic: Set operations, Cartesian products and power sets (§1.2) express union, intersection and difference through membership.

2a: Read four regions in the diagram

Try first: translate each expression using "x is in …" before describing the region in words.



Three overlapping subsets of X. The outlines show the regions named by A, B and C; the membership tests below specify exactly which points each answer keeps.

Answers: see "Exercise 1.2: Sets" under Answers at the end.

2b: Work with even, odd and ordered-pair sets

Try first: separate the integer examples from the ordered-pair example, and check both coordinates of each pair.

Answers: see "Exercise 1.2: Sets" under Answers at the end.

2c: Prove a set identity by membership

Try first: take an arbitrary x and translate both "set difference" steps into and/not conditions.

Answers: see "Exercise 1.2: Sets" under Answers at the end.

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Exercise 1.3: Quantifiers — students and courses

Let S be the student set, C the course set, and $B(s,\ c)$ mean that student s passed course c. The order of "for all" and "there exists" determines who may vary.

Connected topic: Predicates, quantifier order and negation (§1.1) define the domain and the order in which choices are made.

3a: Translate "a bright spark passes everything"

Try first: decide whether the student is chosen before or after the course, and whether one student must pass every course.

Answers: see "Exercise 1.3: Quantifiers — students and courses" under Answers at the end.

3b: Translate "nobody passed the whole programme"

Try first: negate "there is a student who passed every course", then swap the quantifiers as you push the negation inward.

Answers: see "Exercise 1.3: Quantifiers — students and courses" under Answers at the end.

3c: Translate "a course everyone passes"

Try first: identify the one course that must work for every student, then contrast it with "each student has some course".

Answers: see "Exercise 1.3: Quantifiers — students and courses" under Answers at the end.

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Self-Study 1.4: Statements

This review combines statement classification, truth tables, implication language and translating ordinary "unless" sentences.

Connected topic: Statements, truth tables and logical equivalence (§1.1) define the tests used below.

4a: Decide whether each sentence is a statement

Try first: check whether the sentence asserts something with a definite truth value, not whether you personally know the value.

Answers: see "Self-Study 1.4: Statements" under Answers at the end.

4b: Verify two equivalences by truth table

Try first: make one table for the distributive law and a second for De Morgan's law; compare the two target columns row by row.

Answers: see "Self-Study 1.4: Statements" under Answers at the end.

4c: Rewrite "if" and "only if" sentences

Try first: for "X only if Y", write $X \Rightarrow Y$; then substitute the given truth values.

Answers: see "Self-Study 1.4: Statements" under Answers at the end.

4d: Show that a conditional is not a tautology

Try first: to make an implication false, make its antecedent true and its consequent false.

Answers: see "Self-Study 1.4: Statements" under Answers at the end.

4e: Translate "if", "only if" and "unless"

Try first: "B unless G" means that if G is absent, B does not happen. Compare the directions after translating.

Answers: see "Self-Study 1.4: Statements" under Answers at the end.

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Self-Study 1.5: Sets

Use the intervals and point set in the source diagram. Complements here are relative to $\mathbb{R}$, since the exercise defines all three sets as subsets of $\mathbb{R}$.

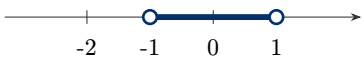
Connected topic: Intervals, complements and power sets (§1.2) define membership and the universe for these operations.

5a: Combine two intervals with a finite set

Try first: write the sets as $A = (-2,\ 1]$, $B = (-1,\ 1)$, and $C = \{-2,\ 0,\ 1\}$; mark the endpoints before combining.



$A = (-2,\ 1]$. The left endpoint is open and the right endpoint is included.



$B = (-1,\ 1)$. Both endpoints are excluded.



$C = \{-2,\ 0,\ 1\}$. The red points mark the three roots of $x(x + 2)(x - 1) = 0$.

Answers: see "Self-Study 1.5: Sets" under Answers at the end.

5b: Form two power sets and test membership

Try first: list subsets by size, and expand the Cartesian product before deciding what its power set contains.

Answers: see "Self-Study 1.5: Sets" under Answers at the end.

5c: Prove a difference identity

Try first: expand "not in $B \setminus C$" and distribute "and" over "or".

Answers: see "Self-Study 1.5: Sets" under Answers at the end.

Self-Study 1.6: Quantifiers and numbers

For each domain, decide whether every quantified claim holds. In these notes, $\mathbb{N} = \{1,\ 2,\ 3,\ \ldots\}$, $\mathbb{Z}$ is the integers, and $\mathbb{R}^+ = (0, \infty)$.

Connected topics: Quantifiers and negation (§1.1) and number systems (§2.1) specify the logic and the three domains.

6a: Evaluate four statements and write their negations

Try first: substitute a test value to disprove a universal claim; to prove one, give a valid witness for every value required.

Answers: see "Self-Study 1.6: Quantifiers and numbers" under Answers at the end.

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Answers

Exercise 1.1: Statements

1a · A definite truth value

1. **i. "z is prime."** The value of z has not been given or quantified. This is an open form: it becomes a statement only after a value is assigned to z (or a quantifier is added). **As written, it is not a statement.**
2. **ii. The claim about Lothar Matthäus and the 2011 Tour de France.** It names a particular person and event, with no free variable. It is a factual declarative sentence, so it has a definite truth value whether or not the reader knows it. **It is a statement.**
3. **iii. "This statement is false."** If it were true, what it says would make it false; if it were false, then what it says would be true. It cannot be assigned a consistent true-or-false value in ordinary two-valued logic. **It is not a logical statement.**
4. **iv. The conditional involving n and the oak trees.** The variable n is free, so different values of n can change the antecedent. The sentence would need a specified n, or a quantifier such as "for every n". **As written, it is not a statement.**

Exercise 1.1: Statements

1b · Truth table for implication and its contrapositive

1. There are two propositions, so there are four input rows. In the last column, the three expressions are linked by biconditionals; that column is true when their truth values agree.

2.

A	B	$A \Rightarrow B$	$\neg B \Rightarrow \neg A$	$\neg(A \wedge \neg B)$	whole chain
T	T	T	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

3. The columns for $A \Rightarrow B$, $\neg B \Rightarrow \neg A$, and $\neg(A \wedge \neg B)$ match in every row. The biconditional chain is therefore true in all four rows: all three formulas are equivalent, and the chain is a tautology.
4. **Check by rules:** $\neg B \Rightarrow \neg A$ is the contrapositive of $A \Rightarrow B$. Also, implication is equivalent to $\neg A \vee B$, and De Morgan's law gives $\neg(A \wedge \neg B) = \neg A \vee B$.

Exercise 1.1: Statements

1c · Translate the clues and eliminate cases

1. Let A, B and C mean that Anton, Bert and Christoph, respectively, are guilty. The three clues become:

$$C \Rightarrow A$$

$$(B \vee C) \Rightarrow \neg A$$

$$(\neg A \vee \neg C) \Rightarrow B$$

2. Suppose C were true. The first clue would make A true. Then $B \vee C$ is true, so the second clue makes $\neg A$ true. That contradicts A; hence C is false.
3. Because $\neg C$ is true, the antecedent $A \vee \neg C$ in the third clue is true. Therefore B is true: Bert is guilty.
4. Now $B \vee C$ is true, so the second clue gives $\neg A$. Anton is innocent, and we already found Christoph innocent.
5. **Conclusion:** B is true while A and C are false. **Bert committed the crime.** Substitution checks all three clues: true, true, and true.

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Exercise 1.2: Sets

2a · Convert each expression to a membership test

1. For i), membership in $A \setminus B$ means "in A and not in B". Intersecting with C adds one more "and":

$$x \in (A, B) \cap C \Leftrightarrow (x \in A) \wedge (x \notin B) \wedge (x \in C)$$

   Keep the part shared by A and C that lies outside B.

2. For ii), union means "or". Keep every point in C, together with any point that is in A but outside B:

$$x \in (A, B) \cup C \Leftrightarrow ((x \in A) \wedge (x \notin B)) \vee (x \in C)$$

3. For iii), a complement keeps the points in X that fail membership in the whole union. De Morgan's law changes "not (P or Q)" to "not P and not Q":

$$((A, B) \cup C)^{\mathrm{c}} = ((A, B)^{\mathrm{c}}) \cap C^{\mathrm{c}} = (A^{\mathrm{c}} \cup B) \cap C^{\mathrm{c}}$$

   So keep points outside C and either outside A or inside B.

4. For iv), first combine the two differences: $A \setminus C$ and $B \setminus C$ together are all of $A \cup B$ with C removed. Complementing adds C back and keeps the points outside both A and B:

$$((A, C) \cup (B, C))^{\mathrm{c}} = ((A \cup B), C)^{\mathrm{c}} = (A \cup B)^{\mathrm{c}} \cup C$$

5. **Check:** in iii) any point inside C is excluded; in iv) every point inside C is included. That opposite treatment is a quick way to catch a misplaced complement.

Made by Thana

Exercise 1.2: Sets

2b · Operations, products and power sets

1. A contains the positive even natural numbers; B contains the positive odd natural numbers. Every positive natural number is exactly one of these, so:

$$A \cup B = \mathbb{N}$$

$$A \cap B = \varnothing$$

$$A, B = A$$

$$B, A = B$$

2. The Cartesian product asks for ordered pairs whose first entry is in the first factor and second entry is in the second:

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

3. Test the two elements of C separately. $(2, 3)$ passes: 2 is even and 3 is odd. $(3, 4)$ fails: 3 is not even and 4 is not odd. Since one element fails, $\boldsymbol{C}$ **is not a subset of** $\boldsymbol{A \times B}$**.**

4. The power set contains every subset of C: the empty set, each one-element subset, and C itself.

$$\mathcal{P}(C) = \{\varnothing, \{(2, 3)\}, \{(3, 4)\}, \{(2, 3), (3, 4)\}\}$$

There are $2^2 = 4$ subsets, matching the two elements of C.

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Exercise 1.2: Sets

2c · Element-by-element proof

1. Take any x. It belongs to the left-hand side exactly when it is in A but not in $A \setminus B$:

$$x \in A, (A, B) \Leftrightarrow (x \in A) \wedge \neg(x \in A, B)$$

2. Expand the inner difference: $x \in A \setminus B$ means $x \in A$ and $x \notin B$. Therefore:

$$(x \in A) \wedge \neg((x \in A) \wedge (x \notin B))$$

3. Because we already know $x \in A$, the negated inner condition reduces to $x \in B$. Thus membership is equivalent to $x \in A$ and $x \in B$:

$$x \in A, (A, B) \Leftrightarrow x \in A \cap B$$

4. This is true for every x, so both sets contain exactly the same elements. Hence $\boldsymbol{A \setminus (A \setminus B) = A \cap B}$.

Exercise 1.3: Quantifiers — students and courses

3a · One student, all courses

1. "There is a student" gives $\exists s$. "Passes everything" says that same student passes every course, giving $\forall c$. The correct choice is **b)** $\boldsymbol{\exists s \in S\ \forall c \in C: B(s, c)}$.

2. Translate the other choices to see the difference:

Choice	Meaning
a) $\forall s\ \exists c\ B(s,c)$	Every student has passed at least one course; it may be a different course for each student.
c) $\exists s\ \exists c\ B(s,c)$	At least one student has passed at least one course.
d) $\forall c\ \exists s\ B(s,c)$	Every course has at least one student who passed it; the student may vary by course.

3. **Check:** $\exists s \forall c$ is stronger than $\forall s \exists c$. The former fixes one student and then asks about every course.

Exercise 1.3: Quantifiers — students and courses

3b · Each student missed a course

1. "Nobody has made it through" means there is no student who passed all courses:

$$\neg(\exists s \in S \, \forall c \in C : B(s, c))$$

2. Negating $\exists$ changes it to $\forall$; negating $\forall$ changes it to $\exists$; the predicate is negated too:

$$\forall s \in S \, \exists c \in C : \neg B(s, c)$$

   So the answer is **d)**: every student has at least one course they have not passed.

3. The other options say:

Choice	Meaning
a) $\exists s \, \forall c \, \neg B(s,c)$	At least one student has passed no courses.
b) $\forall c \, \exists s \, \neg B(s,c)$	Every course has at least one student who has not passed it.
c) $\forall s \, \forall c \, \neg B(s,c)$	No student has passed any course.

4. **Check:** the correct statement allows a student to have passed most courses; it only requires one unpassed course for each student.

Exercise 1.3: Quantifiers — students and courses

3c · One course, all students

1. "There is a course" gives $\exists c$; "every student passes it" gives $\forall s$. The correct choice is **a)** $\exists \boldsymbol{c} \in \boldsymbol{C}$ $\forall \boldsymbol{s} \in \boldsymbol{S}$**:** $\boldsymbol{B(s,\ c)}$**.**

2. The other formulas mean:

Choice	Meaning
b) $\forall c \, \exists s \; B(s,c)$	Every course has some student who passed it.
c) $\forall s \, \exists c \; B(s,c)$	Every student has passed at least one course.
d) $\exists s \, \forall c \; B(s,c)$	Some student has passed every course.

3. **Check:** in a), one course is chosen first and then tested for every student. In d), it is one student tested against every course.

Self-Study 1.4: Statements

4a · Assertion or open judgement?

1. **i. The conditional about a father having a sibling.** It is a complete declarative sentence with a fixed family relation as its subject. It can be true or false, even if the family facts are unknown. **It is a statement.**
2. **ii. "The formula is beautiful."** No formula or precise criterion for "beautiful" is specified. Under the guide's logical definition, the sentence has no definite truth value. **As written, it is not a logical statement.**

Self-Study 1.4: Statements

4b · Distributive law and De Morgan's law

1. For i), evaluate the left side $A \wedge (B \vee C)$ and right side $(A \wedge B) \vee (A \wedge C)$ for all eight assignments:

A	B	C	$A \wedge (B \vee C)$	$(A \wedge B) \vee (A \wedge C)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	F	F
F	T	F	F	F
F	F	T	F	F
F	F	F	F	F

2. The two columns agree in all eight rows, so the distributive equivalence is true for every assignment.
3. For ii), compare both sides of $\neg(A \wedge B) \Leftrightarrow \neg A \vee \neg B$:

A	B	$\neg(A \wedge B)$	$\neg A \vee \neg B$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	T	T

4. These columns also agree in every row. The equivalence is a tautology, which verifies this De Morgan law.

Self-Study 1.4: Statements

4c · Conditional direction and truth value

1. Use C = cats have wings, D = dogs have wings, P = parrots have wings. The given assignment makes all three false.
2. **i. Cats have wings only if dogs have wings.** "Only if" puts the necessary condition second: $C \Rightarrow D$, read "If cats have wings, then dogs have wings." It is true because the antecedent C is false.
3. **ii. Parrots have wings only if dogs have wings.** Formula $P \Rightarrow D$, read "If parrots have wings, then dogs have wings." It is true because P is false.
4. **iii. Parrots have wings if dogs have wings.** The condition after "if" is the antecedent: $D \Rightarrow P$, read "If dogs have wings, then parrots have wings." It is true because D is false.
5. **iv. If cats or dogs have wings, then both cats and dogs have wings.** Formula $(C \vee D) \Rightarrow (C \wedge D)$. It is true because its antecedent is false. **All four sentences are true** for the supplied assignment; a conditional with a false antecedent is true in the truth-table definition.

Self-Study 1.4: Statements

4d · One counterexample is enough

1. Choose A = false and B = true. Then $A \Rightarrow B$ is true (false antecedent), while $B \Rightarrow A$ is false (true antecedent and false consequent).
2. The outer implication is therefore true $\Rightarrow$ false, which is false:

$$(A \Rightarrow B) \Rightarrow (B \Rightarrow A) = \mathrm{T} \Rightarrow \mathrm{F} = \mathrm{F}$$

3. A tautology must be true for every assignment. This single false row proves the formula is **not** a tautology.

Self-Study 1.4: Statements

4e · Mirror conditions

1. Let B mean the mirror breaks, G mean it is made of glass, and W mean it is high on the wall.
2. **i. Breaks if it is made of glass:** glass is sufficient for breaking, so $G \Rightarrow B$.
3. **ii. Breaks only if it is made of glass:** glass is necessary for breaking, so $B \Rightarrow G$.
4. **iii. Unless it is made of glass, it will not break:** if it is not glass, it does not break: $\neg G \Rightarrow \neg B$. By contraposition, this is equivalent to $B \Rightarrow G$.
5. **iv. It is high on the wall, but will not break unless it is glass:** "but" is conjunction, so the full formula is $W \wedge (\neg G \Rightarrow \neg B)$, equivalently $W \wedge (B \Rightarrow G)$.
6. **Equivalent pair:** ii) and iii) express the same conditional. In iv), that conditional is joined to the extra requirement W, so iv) is not generally equivalent to ii).

Self-Study 1.5: Sets

5a · Intervals and roots

1. Factor equation $x(x+2)(x-1)=0$. A product is zero when a factor is zero, so $C=\{-2,\,0,\,1\}$. The interval B sits entirely inside A.
2. $A\cap B$ keeps numbers in both. Since B is contained in A, the answer is $\mathbf{(-1,\,1)}$.
3. $A\cap B\cap C$ asks which listed roots lie in the overlap $(-1,\,1)$. Only 0 does, so the answer is $\mathbf{\{0\}}$.
4. $A\cap(B\cup C)$ starts with $B\cap A=(-1,\,1)$. Of the points in C, 0 is already there, 1 belongs to A, and -2 does not. Thus the union adds only 1: $\mathbf{(-1,\,1]}$.
5. Because $B\subset A$, $A\cup B=A$. Everything in $\mathbb{R}$ outside $(-2,\,1]$ is $\mathbf{(-\infty,\,-2]\cup(1,\,\infty)}$. The point -2 is included in the complement because it is excluded from A; 1 is not.
6. First remove B from A: $A\setminus B=(-2,\,-1]\cup\{1\}$. Then remove C; its point 1 is removed, while -2 was not in the interval to begin with. The result is $\mathbf{(-2,\,-1]}$.
7. **Endpoint check:** -2 never belongs to A; -1 belongs to A but not B; 1 belongs to A and C but not B. These three points catch the most likely endpoint mistakes.

Self-Study 1.5: Sets

5b · Every subset, including the empty set

1. $M_1=\{b,\,a,\,c\}$ has one empty subset, three one-element subsets, three two-element subsets and itself:

$$\mathcal{P}(M_1)=\{\varnothing,\,\{a\},\,\{b\},\,\{c\},\,\{a,b\},\,\{a,c\},\,\{b,c\},\,\{a,b,c\}\}$$

2. Expand the ordered-pair set first:

$$M_2=\{1,\,2\}\times\{1\}=\{(1,\,1),\,(2,\,1)\}$$

3. Its four subsets are:

$$\mathcal{P}(M_2)=\{\varnothing,\,\{(1,\,1)\},\,\{(2,\,1)\},\,\{(1,\,1),\,(2,\,1)\}\}$$

4. **Is $\{1,\,2\}\in\mathcal{P}(M_2)$?** Membership in a power set means being a subset of M_2. But the elements of M_2 are ordered pairs, not the numbers 1 and 2; neither 1 nor 2 is in M_2. Therefore $\{1,\,2\}$ is not a subset of M_2, so $\mathbf{\{1,\,2\}\notin\mathcal{P}(M_2)}$.
5. **Check:** the product has two pairs, so its power set has $2^2=4$ members.

Self-Study 1.5: Sets

5c · Use membership and distributivity

1. Take arbitrary $x \in X$. It belongs to the left side when $x \in A$ and it is not true that x is in B but outside C:

$$x \in A, (B, C) \Leftrightarrow (x \in A) \wedge \neg((x \in B) \wedge (x \notin C))$$

2. By De Morgan's law, the negated bracket says $x \notin B$ or $x \in C$. Distribute the outer "and":

$$(x \in A) \wedge ((x \notin B) \vee (x \in C))$$

$$\Leftrightarrow ((x \in A) \wedge (x \notin B)) \vee ((x \in A) \wedge (x \in C))$$

3. Translate the two terms back into sets: the first is $A \setminus B$; the second is $A \cap C$. Therefore both sides have exactly the same members:

$$A, (B, C) = (A, B) \cup (A \cap C)$$

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Self-Study 1.6: Quantifiers and numbers

6a · Truth values in $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{R}^+$

1. First negate each statement by swapping $\forall$ and $\exists$ and negating the relation:

Statement	Negation
i) $\exists x \in M\ \forall y \in M$: $x < y$	$\forall x \in M\ \exists y \in M$: $x \geq y$
ii) $\forall y \in M\ \exists x \in M$: $x < y$	$\exists y \in M\ \forall x \in M$: $x \geq y$
iii) $\exists x \in M\ \forall y \in M$: $x + y = 1$	$\forall x \in M\ \exists y \in M$: $x + y \neq 1$
iv) $\forall y \in M\ \exists x \in M$: $x + y = 1$	$\exists y \in M\ \forall x \in M$: $x + y \neq 1$

2. **i) False in all three domains.** If x is in M, the universal condition must also hold at $y = x$, which would require $x < x$. That is false for every number.

3. **ii)** For $\mathbb{N}$, take $y = 1$; there is no smaller natural number in this course's convention, so the claim is false. For $\mathbb{Z}$, choose $x = y - 1$ for any integer y; this is always an integer smaller than y, so the claim is true. For $\mathbb{R}^+$, choose $x = y/2$ for each y; it is still positive and is smaller than y. Thus the values are **F, T, T.**

4. **iii) False in all three domains.** The same chosen x would have to satisfy the equation for every y. Testing $y = 1$ forces $x = 0$; testing $y = 2$ forces $x = -1$. One value of x cannot meet both requirements.

5. **iv)** For $\mathbb{N}$, choose $y = 2$: the only possible x is -1, which is not natural, so false. In $\mathbb{Z}$, for every y choose $x = 1 - y$, which is an integer, so true. In $\mathbb{R}^+$, again $y = 2$ would require $x = -1$, not positive, so false. The values are **F, T, F.**

6. Collected results:

Claim	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{R}^+$
i)	F	F	F
ii)	F	T	T
iii)	F	F	F
iv)	F	T	F

7. **Check:** ii) asks whether there is always a smaller member; $\mathbb{Z}$ and $\mathbb{R}^+$ have no least element. iv) asks whether $1 - y$ stays in the same set for every y; this works in $\mathbb{Z}$ but not in the positive domains.

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